BharatLCL: Streamlining LCL Freight for Indian MSMEs

Digital Logistics Marketplace — Jaipur–Mundra Corridor

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PROBLEM STATEMENT

Managing export logistics efficiently is challenging for Indian MSMEs due to fragmented freight consolidation processes. Small exporters struggle with **Full Container Load (FCL)** requirements, resulting in **20–35% dead-freight costs**. They face difficulties finding available LCL container space, verifying logistics partners, tracking shipments, and managing documentation — leading to delays, financial losses, and compromised competitiveness.

OBJECTIVE

BharatLCL aims to simplify export logistics by creating a connected marketplace for exporters, transporters, and administrators.

- **Real-time capacity discovery** for LCL container slots
- **ULIP-based verification** of logistics partners
- **Milestone-based escrow** payments for trust
- **Automated documentation** — shipping bills, invoices, packing lists
- **GPS-based shipment tracking** with live alerts

TECHNOLOGY STACK

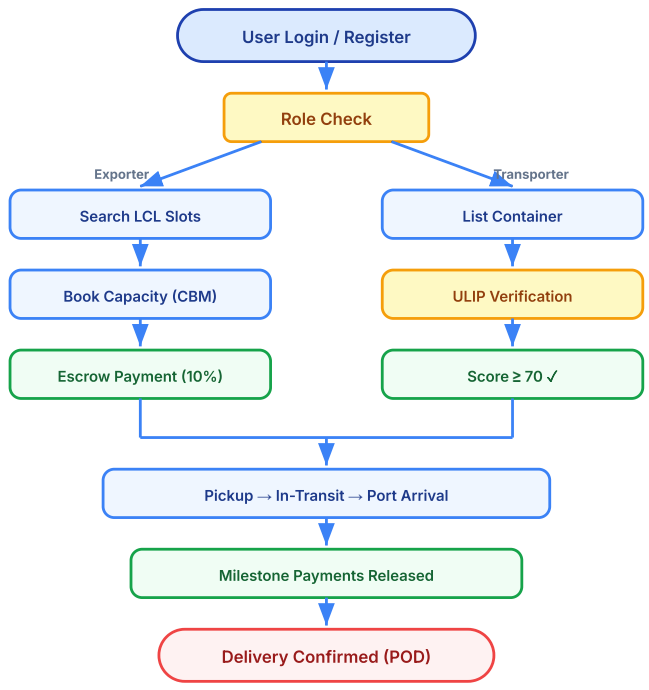
BharatLCL leverages the MERN stack to ensure scalability, performance, and a smooth user experience.

- **React 18 / Vite** — Frontend SPA
- **Node.js / Express** — Backend services and APIs
- **MongoDB Atlas** — Cloud database (Mongoose ODM)
- **JWT + bcrypt** — Authentication and security
- **ULIP APIs** — VAHAN, FASTag, LDB verification
- **Razorpay** — Payment gateway & escrow
- **Vercel** — Frontend deployment & hosting

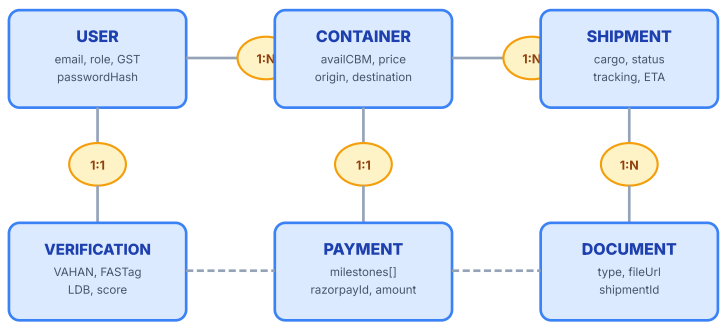
ANALYSIS AND PLANNING

The **Flow Diagram** illustrates user interactions, from capacity discovery to delivery confirmation. The **ER Diagram** represents database relationships between users, containers, shipments, and payments. The **Class Diagram** outlines system components and their interactions, ensuring an efficient and scalable design.

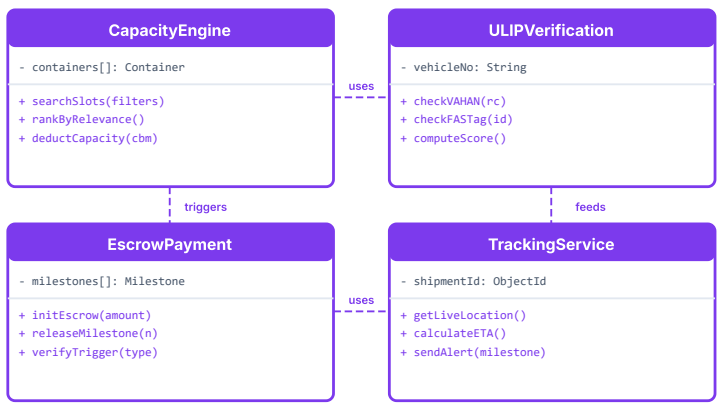
FLOW DIAGRAM



ER DIAGRAM

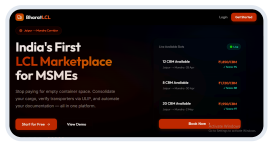


CLASS DIAGRAM



APP INTERFACE & UNIQUE SELLING POINTS

The BharatLCL interface is designed to be intuitive, accessible, and data-rich — ensuring ease of use for both exporters and transporters.



Landing Page & Live Slots

Key Features Overview

Unique Selling Points

Real-Time Discovery
CBM-based LCL capacity search with smart ranking (Price 35%, Date 30%, Fit 20%, Trust 15%)

Govt. ULIP Verified
3-API verification: VAHAN (40pts) + FASTag (30pts) + LDB (30pts). Pass ≥ 70

Escrow Protection
5-milestone payment: 10% → 20% → 20% → 25% → 25%. GPS & FASTag verified

Zero Paperwork
Auto-generated PDFs: Shipping Bills, Commercial Invoices, Packing Lists

CONCLUSION

BharatLCL enhances the export logistics experience by making freight consolidation more organized, accessible, and efficient. It bridges the trust gap through ULIP verification, protects finances via escrow payments, and empowers MSMEs with better control over their export journey.

Future Scope

- **AI-Based Rate Prediction** — Dynamic pricing recommendations using historical freight data
- **Multi-Corridor Expansion** — Extending beyond Jaipur–Mundra to pan-India trade routes
- **Blockchain Provenance** — Immutable shipment records for customs compliance
- **Mobile App** — React Native app for on-the-go booking and tracking